

Regularity-Selected Generic Rank at Least Four in the Piezas Diophantine Sextuple

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Abstract

The Piezas one-parameter rational Diophantine sextuple contains twenty three-element subfamilies. We show that its regularity structure selects six of them in a uniform way and that every selected triple induces an elliptic curve of Mordell–Weil rank at least four over $\mathbb{Q}(t)$. The regular quadruples share the pair $\{e, f\}$; after removing this pair, the remaining incidence graph is the path $c - d - a - b$. The two endpoints and the two distance-two pairs of this path select the six triples

$$ace, acf, bde, bdf, bef, cef.$$

For each selected triple we construct four rational sections explicitly from the Diophantine square witnesses. Independence is proved by specializing all six families at the same good value $t = 12/5$ and certifying positive-definite canonical-height Gram matrices with rigorous interval arithmetic. The result is a generic lower-bound theorem. It does not claim exact rank four or a rank bound for the remaining fourteen triple classes.

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1 Introduction

A rational Diophantine m -tuple is a set of distinct nonzero rational numbers such that the product of any two distinct elements, increased by 1, is a rational square. Rational Diophantine sextuples exist in infinite families [5]. The particularly simple one-parameter sextuple used here was given by Piezas [8] and subsequently studied by Dujella and Kazalicki [4].

A rational Diophantine triple $\{u, v, w\}$ induces an elliptic curve

$$Y^2 = (uX + 1)(vX + 1)(wX + 1).$$

The use of rational Diophantine triples to construct elliptic curves of high rank has a substantial history. Dujella constructed induced families of generic rank at least four [2]; Aguirre, Dujella, and Peral later obtained generic rank at least five [1]; and Dujella and Peral constructed a family of generic rank at least seven [7]. Thus generic rank at least four is not itself the new feature of the present result.

Regular substructures inside rational Diophantine sextuples have also been studied previously. Dujella, Kazalicki, and Petričević constructed infinite families containing regular quadruples and regular quintuples [6]. More recently, Zeng studied regular extensions and compatibility for standard parametric rational Diophantine sextuples over $\mathbb{Q}(t)$ [10].

The contribution here is the link between the internal regularity structure of the Piezas sextuple and the ranks of elliptic curves induced by its three-element subfamilies. The regularity identities themselves are not claimed as new. What is proved here is that their incidence pattern selects six specific triple classes from the twenty available three-element subfamilies, and that each selected class carries four independent rational sections over $\mathbb{Q}(t)$. Thus the theorem gives a regularity-indexed generic rank lower bound for these six induced elliptic-curve families.

The proof has two parts. First, five exact regularity identities produce a small incidence graph that selects six distinguished triples. Second, each selected triple is shown to carry four rational sections. Their independence is certified at the common good specialization

$$t = \frac{12}{5}$$

by positive-definite canonical-height Gram matrices. Only persistence of a section relation under good specialization is used; injectivity of the full specialization homomorphism is not required.

2 The Piezas family and its regularity graph

Let

$$\begin{aligned}
 a &= \frac{(t^2 - 2t - 1)(t^2 + 2t + 3)(3t^2 - 2t + 1)}{4t(t^2 - 1)(t^2 + 2t - 1)}, \\
 b &= \frac{4t(t^2 - 1)(t^2 - 2t - 1)}{(t^2 + 2t - 1)^3}, \\
 c &= \frac{4t(t^2 - 1)(t^2 + 2t - 1)}{(t^2 - 2t - 1)^3}, \\
 d &= \frac{(t^2 + 2t - 1)(t^2 - 2t + 3)(3t^2 + 2t + 1)}{4t(t^2 - 1)(t^2 - 2t - 1)}, \\
 e &= \frac{-t(t^2 + 4t + 1)(t^2 - 4t + 1)}{(t - 1)(t + 1)(t^2 + 2t - 1)(t^2 - 2t - 1)}, \\
 f &= \frac{(t - 1)(t + 1)(3t^2 - 1)(t^2 - 3)}{4t(t^2 + 2t - 1)(t^2 - 2t - 1)}.
 \end{aligned}$$

These functions form the Piezas rational Diophantine sextuple over $\mathbb{Q}(t)$ [4]. Thus for every distinct $i, j \in \{a, b, c, d, e, f\}$ there exists $\rho_{ij} \in \mathbb{Q}(t)$ satisfying

$$\rho_{ij}^2 = ij + 1.$$

We choose one such square root for each unordered pair and keep that choice fixed.

We use the standard algebraic regularity tests [6]. An ordered quadruple (p, q, r, s) is regular when

$$(p - q - r + s)^2 = 4(ps + 1)(qr + 1), \quad (1)$$

and an ordered quintuple (p, q, r, s, z) is regular when

$$(pqrsz + 2pqr + p + q + r - s - z)^2 = 4(pq + 1)(pr + 1)(qr + 1)(sz + 1). \quad (2)$$

These identities are convenient here because they can be checked directly in $\mathbb{Q}(t)$; no specialization is involved.

Proposition 1 (Regularity pattern). *For the Piezas family, the identities (1) and (2) hold identically in $\mathbb{Q}(t)$ for the ordered quadruples*

$$(a, b, e, f), \quad (a, d, e, f), \quad (c, d, e, f),$$

and the ordered quintuples

$$(a, b, c, d, e), \quad (a, b, c, d, f).$$

Proof. The claim is an exact rational-function calculation. Substitute the six displayed Piezas functions into the left side minus the right side of (1) for

$$(a, b, e, f), \quad (a, d, e, f), \quad (c, d, e, f),$$

and into the corresponding residual of (2) for

$$(a, b, c, d, e), \quad (a, b, c, d, f).$$

After cancellation in the fraction field $\mathbb{Q}(t)$, all five residuals are zero. □

For the combinatorial argument, we now forget the ordering and retain only the underlying label sets. The three regular quadruples share the pair

$$\{e, f\}.$$

Removing that pair leaves the two-element sets

$$\{a, b\}, \quad \{a, d\}, \quad \{c, d\},$$

which we regard as edges on the vertex set $\{a, b, c, d\}$. Thus the resulting graph is the path

$$c - d - a - b. \tag{3}$$

Its endpoints are b, c , and its distance-two pairs are ac, bd .

Definition 2 (Regularity-selected triples). The regularity-selected triples are obtained in either of two ways:

1. take the special pair $\{e, f\}$ together with an endpoint of (3);
2. take one element of $\{e, f\}$ together with a distance-two pair of (3).

Thus

$$\mathcal{S}_P = \{ace, acf, bde, bdf, bef, cef\}. \tag{4}$$

There are exactly six such triples: two of the first type and four of the second. This is a selection rule coming from the regularity graph. It does not by itself say anything about the ranks of the other fourteen triple classes.

3 Four sections on each selected curve

For a Diophantine triple $T = \{u, v, w\}$, we use the standard equivalent model for elliptic curves induced by Diophantine triples [3],

$$E_T : \quad y^2 = (x + uv)(x + uw)(x + vw). \tag{5}$$

It is obtained from

$$Y^2 = (uX + 1)(vX + 1)(wX + 1)$$

by the change of variables

$$x = uvwX, \quad y = uvwY.$$

In the six cases considered below, u, v, w are nonzero and pairwise distinct as rational functions, so (5) is nonsingular over $\mathbb{Q}(t)$.

Write

$$r = \rho_{uv}, \quad s = \rho_{uw}, \quad q = \rho_{vw},$$

so that

$$r^2 = uv + 1, \quad s^2 = uw + 1, \quad q^2 = vw + 1.$$

The standard Diophantine-triple construction gives the rational point [3]

$$P_T = (0, uvw). \tag{6}$$

The same construction gives

$$R_T = (rs + rq + sq + 1, (r + s)(r + q)(s + q)). \quad (7)$$

Direct substitution using

$$r^2 = uv + 1, \quad s^2 = uw + 1, \quad q^2 = vw + 1$$

verifies that

$$R_T \in E_T(\mathbb{Q}(t)).$$

Every fourth element z from the ambient sextuple gives another rational point

$$E_z = (uvwz, uvw\rho_{uz}\rho_{vz}\rho_{wz}) \in E_T(\mathbb{Q}(t)). \quad (8)$$

Indeed, substitution into (5) gives

$$(uz + 1)(vz + 1)(wz + 1) = \rho_{uz}^2 \rho_{vz}^2 \rho_{wz}^2.$$

For the six selected triples we use the following four sections.

T	four sections
ace	P_T, R_T, E_d, E_f
acf	P_T, R_T, E_d, E_e
bde	P_T, R_T, E_c, E_f
bdf	P_T, R_T, E_c, E_e
bef	P_T, R_T, E_c, E_d
cef	P_T, R_T, E_a, E_b

Proposition 3 (Section identities). *For every $T \in \mathcal{S}_P$, the four sections listed above lie in $E_T(\mathbb{Q}(t))$.*

Proof. The point P_T satisfies (5) immediately.

For R_T , substitute (7) into (5) and use

$$r^2 = uv + 1, \quad s^2 = uw + 1, \quad q^2 = vw + 1.$$

The resulting residual is zero.

For E_z , equation (8) and the identities

$$\rho_{uz}^2 = uz + 1, \quad \rho_{vz}^2 = vz + 1, \quad \rho_{wz}^2 = wz + 1$$

give the curve equation directly.

Thus all four listed sections have coordinates in $\mathbb{Q}(t)$ and satisfy (5) identically. $\square$

4 Independence from one good specialization

It remains to prove that the four sections chosen for each of the six selected families are independent over $\mathbb{Q}(t)$. We do this at one common good specialization.

Lemma 4 (Relation persistence). *Let $E/\mathbb{Q}(t)$ be an elliptic curve and let $Q_1, \dots, Q_m \in E(\mathbb{Q}(t))$. Suppose that $t_0 \in \mathbb{Q}$ is a value at which E specializes to a nonsingular elliptic curve and all Q_i specialize to finite rational points. If*

$$Q_1(t_0), \dots, Q_m(t_0)$$

are independent in $E_{t_0}(\mathbb{Q})$, then

$$Q_1, \dots, Q_m$$

are independent in $E(\mathbb{Q}(t))$.

Proof. Any integral relation

$$n_1 Q_1 + \dots + n_m Q_m = O$$

among the generic sections specializes at a good value t_0 to

$$n_1 Q_1(t_0) + \dots + n_m Q_m(t_0) = O.$$

If the specialized points are independent, every n_i must be zero. □

We use the common specialization

$$t_0 = \frac{12}{5}.$$

For each

$$T \in \{ace, acf, bde, bdf, bef, cef\},$$

the specialized curve is nonsingular and the four sections listed in Section 3 specialize to finite rational points.

Their exact coordinates and the specialized curves are recorded in Appendix A. Appendix B gives rigorous interval enclosures for the complete 4×4 canonical-height Gram matrix of each four-point set. The certification method is described in Appendix C.

For points $Q_1, \dots, Q_4$, write

$$G_T = (\langle Q_i, Q_j \rangle)_{1 \leq i, j \leq 4},$$

where $\langle \cdot, \cdot \rangle$ is the Néron–Tate pairing. The interval-Cholesky certificate in Appendix B has four strictly positive pivot intervals for every selected class. Hence each exact matrix G_T is positive definite.

Proposition 5 (Specialized independence). *At $t = 12/5$, the four selected points are independent on each of the six curves*

$$E_{ace}, E_{acf}, E_{bde}, E_{bdf}, E_{bef}, E_{cef}.$$

Proof. For each curve, Appendix B certifies that the exact canonical-height Gram matrix G_T is positive definite. Therefore, for every nonzero $n \in \mathbb{Z}^4$,

$$\hat{h}\left(\sum_{i=1}^4 n_i Q_i\right) = n^\top G_T n > 0.$$

Thus no nontrivial integral relation exists among the four specialized points. □

By Lemma 4, the corresponding four sections are therefore independent over $\mathbb{Q}(t)$.

This argument uses specialization only to rule out relations among the four explicitly constructed sections. It does not require injectivity of the full specialization homomorphism.

5 Main theorem

The regularity graph selects six triple classes from the Piezas sextuple. For each selected class, Section 3 constructs four rational sections, and Section 4 proves their independence by specialization.

Theorem 6 (Regularity-selected generic rank). *Let $a, b, c, d, e, f \in \mathbb{Q}(t)$ be the Piezas sextuple defined above. For every*

$$T \in \{ace, acf, bde, bdf, bef, cef\},$$

let $E_T/\mathbb{Q}(t)$ be the elliptic curve induced by T . Then

$$\text{rank } E_T(\mathbb{Q}(t)) \geq 4.$$

More precisely, the four sections listed in Section 3 are independent in $E_T(\mathbb{Q}(t))$.

Proof. Proposition 3 gives four rational sections on each selected curve. Proposition 5 proves that these four sections specialize to independent rational points at the common good value

$$t = \frac{12}{5}.$$

By Lemma 4, any integral relation among the generic sections would specialize to the same relation among those four specialized points. No such nontrivial relation exists. Therefore the four generic sections are independent, and

$$\text{rank } E_T(\mathbb{Q}(t)) \geq 4$$

for every selected triple T . □

Corollary 7. *The regularity pattern determined by the special pair $\{e, f\}$ and the path*

$$c - d - a - b$$

selects six induced elliptic-curve families, each with generic rank at least four.

Scope. The theorem proves a generic rank lower bound for the six selected families. It does not prove that their generic rank is exactly four. It gives no rank upper bound for the remaining fourteen triple classes, and it does not assert that the regularity-selection condition is necessary for generic rank at least four.

6 Discussion

The result here is structural. The six $\text{rank} \geq 4$ families are not identified by a rank search alone. They are selected first by the regularity pattern of the parent sextuple, and the rank lower bound is proved afterward. That order matters.

The three regular quadruples share the pair $\{e, f\}$. Removing that pair leaves the path

$$c - d - a - b.$$

Its endpoints and distance-two pairs select

$$ace, acf, bde, bdf, bef, cef.$$

For each of these six selected triples, the theorem proves four independent rational sections over $\mathbb{Q}(t)$.

The height calculation is used only to certify independence at one good specialization. It does not choose the six families. The selection comes from the regularity structure itself. This separates the structural part of the argument from the rank certification step.

This proves the forward direction only. The six selected families have generic rank at least four. The theorem does not show that their generic rank is exactly four, and it does not show that the other fourteen triple classes have generic rank at most three. It also does not show that the regularity rule is necessary for generic rank at least four. Additional independent sections may exist on either side of the split.

The next test is to leave the Piezas family. If another parametric rational Diophantine sextuple has a different internal regularity pattern, does that pattern select a different rank stratification? If so, then the induced elliptic curves retain information from the parent Diophantine structure that is lost when the curves are studied one at a time. The present theorem proves one concrete case of that mechanism.

Use of language models

Large language models were used extensively for mathematical derivation, literature review, computational cross-checking, and manuscript editing. The research direction, structural ideas, and computational implementation were developed by the author. The author independently reviewed the code and verified the reported computational outputs, but does not claim independent expert verification of every mathematical derivation. All mathematical claims remain subject to independent review.

A Exact specialization data

This appendix records the exact rational data used in the specialization argument of Proposition 5. No floating-point arithmetic enters the construction of the curves or points.

We specialize throughout at

$$t_0 = \frac{12}{5}.$$

At this value the Piezas sextuple becomes

$$\begin{aligned} a &= -\frac{38081}{2275280}, & b &= -\frac{28560}{13651919}, & c &= -6825840, \\ d &= -\frac{4550799}{9520}, & e &= -\frac{1742340}{28441}, & f &= -\frac{1113959}{19120}. \end{aligned}$$

All fifteen Diophantine square conditions are exact. With the positive square roots chosen at this specialization, we have

$$\rho_{ij}^2 = ij + 1$$

with

ij	ρ_{ij}	ij	ρ_{ij}	ij	ρ_{ij}
ab	$\frac{57122}{57121}$	ac	338	ad	$\frac{28561}{9520}$
ae	$\frac{80951}{56882}$	af	$\frac{26871}{19120}$	bc	$\frac{28561}{239}$
bd	$\frac{338}{239}$	be	$\frac{60671}{57121}$	bf	$\frac{60502}{57121}$
cd	57122	ce	20449	cf	19942
de	$\frac{40729}{238}$	df	$\frac{13351}{80}$	ef	$\frac{28561}{478}$

For a specialized triple $T = \{u, v, w\}$ we use the model

$$E_T : \quad y^2 = (x + uv)(x + uw)(x + vw).$$

The four points recorded below are precisely the specializations of the four rational sections used in Theorem 6. Direct substitution verifies each point on its stated curve.

The class ace

The specialized curve is

$$E_{ace} : \quad y^2 = (x + 114243) \left(x + \frac{3317502477}{3235561924} \right) (x + 418161600).$$

The four points are

$$\begin{aligned} P &= \left(0, -\frac{199050148620}{28441} \right), \\ R &= \left(\frac{394837631403}{56882}, \frac{466858506265862215401}{3235561924} \right), \\ E_d &= \left(\frac{189505693993671}{56644}, -\frac{330661077157423728610479}{1611012004} \right), \\ E_f &= \left(\frac{93165422061591}{228484}, -\frac{76160268231819882592131}{6498313444} \right). \end{aligned}$$

The class acf

The specialized curve is

$$E_{acf} : y^2 = (x + 114243) \left(x + \frac{356476241}{365574400} \right) (x + 397683363).$$

The four points are

$$\begin{aligned} P &= \left(0, -\frac{127262018037}{19120} \right), \\ R &= \left(\frac{3235533363}{478}, \frac{1254589890156810987}{9139360} \right), \\ E_d &= \left(\frac{20362992314643}{6400}, -\frac{11646956640529776327333}{61184000} \right), \\ E_e &= \left(\frac{93165422061591}{228484}, -\frac{50561453069318051543397}{4368614080} \right). \end{aligned}$$

The class bde

The specialized curve is

$$E_{bde} : y^2 = \left(x + \frac{57123}{57121} \right) \left(x + \frac{418161600}{3262808641} \right) \left(x + \frac{1658794797}{56644} \right).$$

The four points are

$$\begin{aligned} P &= \left(0, -\frac{99527687820}{1624578361} \right), \\ R &= \left(\frac{1385065952289}{3249156722}, \frac{3249936130500360056925}{44171629305932156} \right), \\ E_c &= \left(\frac{23886645076800}{57121}, -\frac{3320418339500155352161560}{388274228279} \right), \\ E_f &= \left(\frac{46583934284151}{13051234564}, -\frac{57403828718851406957451}{88714448774099036} \right). \end{aligned}$$

The class bdf

The specialized curve is

$$E_{bdf} : y^2 = \left(x + \frac{57123}{57121} \right) \left(x + \frac{397683363}{3262808641} \right) \left(x + \frac{178242801}{6400} \right).$$

The four points are

$$\begin{aligned} P &= \left(0, -\frac{63632679957}{1092153520} \right), \\ R &= \left(\frac{113387422269}{273038380}, \frac{87230163507296259879}{1247698024318400} \right), \\ E_c &= \left(\frac{22716866744649}{57121}, -\frac{517566012179744500059087}{65256172820} \right), \\ E_e &= \left(\frac{46583934284151}{13051234564}, -\frac{37739102057967529702197}{59639965562419520} \right). \end{aligned}$$

The class bef

The specialized curve is

$$E_{bef} : y^2 = \left(x + \frac{418161600}{3262808641}\right) \left(x + \frac{397683363}{3262808641}\right) \left(x + \frac{815502237}{228484}\right).$$

The four points are

$$\begin{aligned} P &= \left(0, -\frac{5822685972180}{779811265199}\right), \\ R &= \left(\frac{841003521633}{6525617282}, \frac{24470005333732101}{3119245060796}\right), \\ E_c &= \left(\frac{166295911365460800}{3262808641}, -\frac{67816842351097230485934840}{186374892382561}\right), \\ E_d &= \left(\frac{46583934284151}{13051234564}, -\frac{224828135616926173761}{745499569530244}\right). \end{aligned}$$

The class cef

The specialized curve is

$$E_{cef} : y^2 = (x + 418161600)(x + 397683363) \left(x + \frac{815502237}{228484}\right).$$

The four points are

$$\begin{aligned} P &= \left(0, -\frac{5822685972180}{239}\right), \\ R &= \left(\frac{820414727}{2}, \frac{15839801271764899}{956}\right), \\ E_a &= \left(\frac{93165422061591}{228484}, -\frac{899371586570760771039}{54607676}\right), \\ E_b &= \left(\frac{166295911365460800}{3262808641}, -\frac{610446576036399933920249160}{186374892382561}\right). \end{aligned}$$

The data above are exact. In particular, the independence calculation in Appendix B begins from rational curves and rational points; the interval arithmetic enters only when enclosing canonical heights.

B Certified canonical-height Gram data

This appendix records the complete interval Gram enclosures used to prove Proposition 5. The canonical-height pairing is

$$\langle Q_i, Q_j \rangle = \frac{1}{2} \left(\widehat{h}(Q_i + Q_j) - \widehat{h}(Q_i) - \widehat{h}(Q_j) \right).$$

For each selected curve, let

$$G_T = (\langle Q_i, Q_j \rangle)_{1 \leq i, j \leq 4}.$$

Every entry below is an interval containing the corresponding exact entry of G_T . The displayed decimal intervals have been enlarged outward to six decimal places from the intervals used in the proof. Thus the printed intervals remain rigorous enclosures.

The matrices are listed in the exact point order used by the fixed-order interval Cholesky calculation. After each matrix we give the four resulting pivot intervals. Strict positivity of the lower endpoint of every pivot proves positive definiteness of the exact Gram matrix.

The class *ace*

The point order is

$$(P, R, E_f, E_d).$$

$$G_{ace} \in \begin{pmatrix} [6.773169, 6.822800] & [0.110643, 0.185089] & [-1.527812, -1.453366] & [-0.752920, -0.678475] \\ [0.110643, 0.185089] & [1.833062, 1.882692] & [0.031173, 0.105618] & [-1.495576, -1.421131] \\ [-1.527812, -1.453366] & [0.031173, 0.105618] & [6.838918, 6.888548] & [-0.170711, -0.096265] \\ [-0.752920, -0.678475] & [-1.495576, -1.421131] & [-0.170711, -0.096265] & [5.712859, 5.762489] \end{pmatrix}.$$

The Cholesky pivots satisfy

$$\begin{aligned} p_1 &\in [6.773169, 6.822800], & p_2 &\in [1.828004, 1.880898], \\ p_3 &\in [6.482412, 6.577364], & p_4 &\in [4.409636, 4.649906]. \end{aligned}$$

The class *acf*

The point order is

$$(P, R, E_e, E_d).$$

$$G_{acf} \in \begin{pmatrix} [6.609255, 6.656429] & [0.258772, 0.329533] & [1.759208, 1.829969] & [0.679860, 0.750620] \\ [0.258772, 0.329533] & [1.614795, 1.661969] & [0.054934, 0.125694] & [1.143375, 1.214136] \\ [1.759208, 1.829969] & [0.054934, 0.125694] & [6.646619, 6.693793] & [-0.178178, -0.107417] \\ [0.679860, 0.750620] & [1.143375, 1.214136] & [-0.178178, -0.107417] & [5.156196, 5.203370] \end{pmatrix}.$$

The Cholesky pivots satisfy

$$\begin{aligned} p_1 &\in [6.609255, 6.656429], & p_2 &\in [1.598365, 1.651909], \\ p_3 &\in [6.137884, 6.228857], & p_4 &\in [4.158465, 4.382636]. \end{aligned}$$

The class *bde*

The point order is

$$(P, R, E_f, E_c).$$

$$G_{bde} \in \begin{pmatrix} [6.820699, 6.872219] & [0.002413, 0.079692] & [-1.305068, -1.227789] & [-0.441021, -0.363742] \\ [0.002413, 0.079692] & [1.885343, 1.936863] & [-0.561283, -0.484004] & [-2.209275, -2.131997] \\ [-1.305068, -1.227789] & [-0.561283, -0.484004] & [6.557134, 6.608653] & [0.797995, 0.875274] \\ [-0.441021, -0.363742] & [-2.209275, -2.131997] & [0.797995, 0.875274] & [6.505074, 6.556594] \end{pmatrix}.$$

The Cholesky pivots satisfy

$$\begin{aligned} p_1 &\in [6.820699, 6.872219], & p_2 &\in [1.884412, 1.936862], \\ p_3 &\in [6.140499, 6.275849], & p_4 &\in [3.872491, 4.201377]. \end{aligned}$$

The class bdf

The point order is

$$(P, R, E_e, E_c).$$

$$G_{bdf} \in \begin{pmatrix} [6.933809, 6.981345] & [0.152815, 0.224118] & [-1.880476, -1.809174] & [-0.158499, -0.087196] \\ [0.152815, 0.224118] & [1.668454, 1.715990] & [-0.728891, -0.657589] & [1.967046, 2.038349] \\ [-1.880476, -1.809174] & [-0.728891, -0.657589] & [6.637595, 6.685131] & [-0.398411, -0.327108] \\ [-0.158499, -0.087196] & [1.967046, 2.038349] & [-0.398411, -0.327108] & [6.450297, 6.497832] \end{pmatrix}.$$

The Cholesky pivots satisfy

$$\begin{aligned} p_1 &\in [6.933809, 6.981345], & p_2 &\in [1.661210, 1.712645], \\ p_3 &\in [5.841594, 6.008324], & p_4 &\in [3.890490, 4.223149]. \end{aligned}$$

The class bef

The point order is

$$(P, R, E_d, E_c).$$

$$G_{bef} \in \begin{pmatrix} [6.561903, 6.598280] & [1.678559, 1.733124] & [-0.471727, -0.417162] & [-0.145952, -0.091387] \\ [1.678559, 1.733124] & [1.888092, 1.924469] & [-0.598681, -0.544116] & [-0.030757, 0.023808] \\ [-0.471727, -0.417162] & [-0.598681, -0.544116] & [4.841693, 4.878070] & [0.627790, 0.682355] \\ [-0.145952, -0.091387] & [-0.030757, 0.023808] & [0.627790, 0.682355] & [9.228621, 9.264998] \end{pmatrix}.$$

The Cholesky pivots satisfy

$$\begin{aligned} p_1 &\in [6.561903, 6.598280], & p_2 &\in [1.430341, 1.497454], \\ p_3 &\in [4.638163, 4.734163], & p_4 &\in [9.117599, 9.183914]. \end{aligned}$$

The class cef

The point order is

$$(P, E_b, E_a, R).$$

$$G_{cef} \in \begin{pmatrix} [6.662849, 6.700899] & [-0.307507, -0.250432] & [0.243157, 0.300232] & [1.623072, 1.680146] \\ [-0.307507, -0.250432] & [10.718389, 10.756439] & [0.246104, 0.303178] & [-0.030764, 0.026310] \\ [0.243157, 0.300232] & [0.246104, 0.303178] & [5.876537, 5.914587] & [-0.032975, 0.024099] \\ [1.623072, 1.680146] & [-0.030764, 0.026310] & [-0.032975, 0.024099] & [1.720959, 1.759009] \end{pmatrix}.$$

The Cholesky pivots satisfy

$$\begin{aligned} p_1 &\in [6.662849, 6.700899], & p_2 &\in [10.704197, 10.747079], \\ p_3 &\in [5.853619, 5.899704], & p_4 &\in [1.294142, 1.365576]. \end{aligned}$$

Proposition 8 (Verification of specialized independence). *For each*

$$T \in \{ace, acf, bde, bdf, bef, cef\},$$

the four rational points recorded in Appendix A are independent in $E_T(\mathbb{Q})$ at $t = 12/5$.

Proof. For every T , each lower endpoint of each fixed-order interval-Cholesky pivot is strictly positive. Hence every real symmetric matrix contained in the displayed interval enclosure is positive definite. The exact canonical-height Gram matrix is one such matrix. Therefore, for every nonzero $n \in \mathbb{Z}^4$,

$$\hat{h}\left(\sum_{i=1}^4 n_i Q_i\right) = n^\top G_T n > 0.$$

Thus no nontrivial integral relation exists among the four points. $\square$

C Canonical-height certification method

This appendix describes the rigorous height enclosure used in Appendix B. The purpose is to identify the numerical step that carries the independence proof and to state the bounds used there.

C.1 Short-Weierstrass certification model

Write a specialized curve in generalized Weierstrass form

$$E : \quad y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6.$$

Let $b_2, b_4, b_6, b_8, c_4, c_6$ denote the standard Weierstrass invariants. The exact change of variables

$$X = 36x + 3b_2, \quad Y = 216y + 108a_1x + 108a_3$$

maps E to the short Weierstrass model

$$E^{\text{sh}} : \quad Y^2 = X^3 + AX + B,$$

where

$$A = -27c_4, \quad B = -54c_6.$$

This model is used only for the height certificate. It is $\mathbb{Q}$ -isomorphic to the original curve.

For a projective x -coordinate $(X : Z)$, define

$$h_x(X : Z) = \log \max\{|X|, |Z|\},$$

after replacing (X, Z) by a primitive integral pair.

On the short model, duplication of the x -coordinate is represented by the homogeneous quartic forms

$$F(X, Z) = X^4 - 2AX^2Z^2 - 8BXZ^3 + A^2Z^4, \tag{9}$$

$$G(X, Z) = 4X^3Z + 4AXZ^3 + 4BZ^4. \tag{10}$$

After clearing denominators and removing common content, F and G are taken to be primitive integral forms.

C.2 A uniform duplication-height bound

The height error is controlled by exact Bézout certificates for the two affine charts of the duplication map.

Define

$$f_x(T) = F(1, T), \quad g_x(T) = G(1, T),$$

and

$$f_z(T) = F(T, 1), \quad g_z(T) = G(T, 1).$$

Thus the x -chart is used when $|X| \geq |Z|$, with $T = Z/X$, and the z -chart is used when $|Z| \geq |X|$, with $T = X/Z$.

Exact extended Euclidean calculations give integer polynomials s_x, t_x, s_z, t_z and nonzero integers c_x, c_z satisfying

$$s_x f_x + t_x g_x = c_x, \quad s_z f_z + t_z g_z = c_z.$$

Put

$$K_x = \|s_x\|_1 + \|t_x\|_1, \quad K_z = \|s_z\|_1 + \|t_z\|_1,$$

and

$$C_{\text{up}} = \max\{1, \|F\|_1, \|G\|_1\}.$$

The certification constant is

$$C_E = \max\{1, C_{\text{up}}, K_z|c_x|, K_x|c_z|\}. \quad (11)$$

All quantities entering (11) are obtained with exact integer arithmetic.

Lemma 9 (Uniform duplication-height bound). *For every rational point Q on the short certification model,*

$$|h_x(2Q) - 4h_x(Q)| \leq \log C_E. \quad (12)$$

Proof. Let (X, Z) be the primitive integral projective x -coordinate of Q , and put

$$M = \max\{|X|, |Z|\}.$$

Write

$$F_Q = F(X, Z), \quad G_Q = G(X, Z), \quad d = \gcd(|F_Q|, |G_Q|).$$

The duplication forms have no common projective zero on a smooth elliptic curve, so F_Q and G_Q are not both zero. The primitive projective x -coordinate of $2Q$ is therefore

$$\left(\frac{F_Q}{d} : \frac{G_Q}{d} \right).$$

For the upper bound, the coefficient norms give

$$|F_Q| \leq \|F\|_1 M^4, \quad |G_Q| \leq \|G\|_1 M^4.$$

Since $d \geq 1$,

$$\max \left\{ \left| \frac{F_Q}{d} \right|, \left| \frac{G_Q}{d} \right| \right\} \leq C_{\text{up}} M^4 \leq C_E M^4.$$

Taking logarithms gives

$$h_x(2Q) - 4h_x(Q) \leq \log C_E.$$

For the reverse bound, first control the common divisor d . Homogenizing the two Bézout identities gives identities of the form

$$S_x(X, Z)F(X, Z) + T_x(X, Z)G(X, Z) = c_x X^{N_x},$$

and

$$S_z(X, Z)F(X, Z) + T_z(X, Z)G(X, Z) = c_z Z^{N_z},$$

for suitable homogeneous integer polynomials S_x, T_x, S_z, T_z and positive integers N_x, N_z . Since d divides both F_Q and G_Q ,

$$d \mid c_x X^{N_x}, \quad d \mid c_z Z^{N_z}.$$

Because (X, Z) is primitive,

$$\gcd(X, Z) = 1.$$

Prime by prime, the two divisibilities therefore imply

$$d \mid c_x c_z,$$

and hence

$$d \leq |c_x c_z|.$$

Suppose first that $|X| = M$. Then $|Z/X| \leq 1$. Evaluating the x -chart Bézout identity at $T = Z/X$ gives

$$|c_x| \leq (|s_x(Z/X)| + |t_x(Z/X)|) \max \{|f_x(Z/X)|, |g_x(Z/X)|\}.$$

Since $|Z/X| \leq 1$,

$$|s_x(Z/X)| + |t_x(Z/X)| \leq K_x.$$

Using

$$F_Q = X^4 f_x(Z/X), \quad G_Q = X^4 g_x(Z/X),$$

we obtain

$$|c_x| M^4 \leq K_x \max\{|F_Q|, |G_Q|\}.$$

After division by d and using $d \leq |c_x c_z|$,

$$\max \left\{ \left| \frac{F_Q}{d} \right|, \left| \frac{G_Q}{d} \right| \right\} \geq \frac{M^4}{K_x |c_z|}.$$

If instead $|Z| = M$, the same argument on the reciprocal chart gives

$$\max \left\{ \left| \frac{F_Q}{d} \right|, \left| \frac{G_Q}{d} \right| \right\} \geq \frac{M^4}{K_z |c_x|}.$$

Therefore, in both cases,

$$M^4 \leq \max\{K_z |c_x|, K_x |c_z|\} \max \left\{ \left| \frac{F_Q}{d} \right|, \left| \frac{G_Q}{d} \right| \right\}.$$

By (11),

$$4h_x(Q) - h_x(2Q) \leq \log C_E.$$

Combining the two bounds proves (12). □

We use the standard Néron–Tate canonical height and associated bilinear pairing [9]. With the normalization used here,

$$\widehat{h}(Q) = \lim_{n \rightarrow \infty} \frac{h_x(2^n Q)}{2 \cdot 4^n}.$$

Applying Lemma 9 along the duplication tower gives

$$\left| \widehat{h}(Q) - \frac{h_x(2^n Q)}{2 \cdot 4^n} \right| \leq \frac{\log C_E}{6 \cdot 4^n}. \quad (13)$$

Indeed,

$$\begin{aligned} \left| \widehat{h}(Q) - \frac{h_x(2^n Q)}{2 \cdot 4^n} \right| &\leq \sum_{k=n}^{\infty} \frac{|h_x(2^{k+1} Q) - 4h_x(2^k Q)|}{2 \cdot 4^{k+1}} \\ &\leq \frac{\log C_E}{2} \sum_{k=n}^{\infty} \frac{1}{4^{k+1}} \\ &= \frac{\log C_E}{6 \cdot 4^n}. \end{aligned}$$

For all six specialized curves, the maximum in (11) is attained by the same term:

$$C_E = K_x |c_z|.$$

The resulting exact integers are large, ranging from 195 to 275 decimal digits. Their sizes are:

T	term attaining C_E	decimal digits of C_E
ace	$K_x c_z $	265
acf	$K_x c_z $	252
bde	$K_x c_z $	275
bdf	$K_x c_z $	254
bef	$K_x c_z $	195
cef	$K_x c_z $	204

The proof uses $n = 6$ throughout. For reference, the curve constants give the following outward-rounded upper bounds:

T	upper bound for $\log C_E$	upper bound for $\varepsilon_6 = \frac{\log C_E}{6 \cdot 4^6}$
ace	609.849026	0.024815
acf	579.662590	0.023587
bde	633.059132	0.025760
bdf	584.105307	0.023768
bef	446.987750	0.018188
cef	467.544700	0.019025

Thus, if

$$m_n(Q) = \frac{h_x(2^n Q)}{2 \cdot 4^n},$$

then the certified height interval is

$$I_h(Q) = [\max\{0, m_n(Q) - \varepsilon_n\}, m_n(Q) + \varepsilon_n], \quad \varepsilon_n = \frac{\log C_E}{6 \cdot 4^n}. \quad (14)$$

The lower endpoint may be replaced by zero because the canonical height is nonnegative.

The duplication tower itself is computed in exact rational/projective arithmetic. Floating-point arithmetic is used only to represent interval endpoints, and every elementary interval operation is rounded outward.

C.3 Pairing intervals

For rational points Q_i, Q_j , polarization gives

$$\langle Q_i, Q_j \rangle = \frac{\widehat{h}(Q_i + Q_j) - \widehat{h}(Q_i) - \widehat{h}(Q_j)}{2}.$$

If

$$I_h(Q_i) = [a_i, b_i], \quad I_h(Q_j) = [a_j, b_j], \quad I_h(Q_i + Q_j) = [a_{ij}, b_{ij}],$$

then an enclosing interval for the pairing is

$$I_{ij} = \left[\frac{a_{ij} - b_i - b_j}{2}, \frac{b_{ij} - a_i - a_j}{2} \right]. \quad (15)$$

The diagonal entries are the intervals $I_h(Q_i)$. Applying (14) and (15) to the four points on each curve produces the six interval matrices displayed in Appendix B.

C.4 Interval Cholesky certificate

Let

$$\mathcal{G} = (I_{ij})$$

be one of the symmetric interval matrices in Appendix B. Fixed-order interval Cholesky is performed with outward-rounded interval arithmetic.

At step i , suppose the preceding interval entries of the lower-triangular factor have been constructed. The next squared diagonal entry is enclosed by

$$D_i = I_{ii} - \sum_{m < i} L_{im}^2.$$

If

$$\inf D_i > 0,$$

then $\sqrt{D_i}$ is defined and has strictly positive lower endpoint. The off-diagonal entries in column i are then enclosed by

$$L_{ji} = \frac{I_{ji} - \sum_{m < i} L_{jm} L_{im}}{\sqrt{D_i}}, \quad j > i.$$

Lemma 10 (Interval Cholesky certificate). *Suppose the fixed-order interval Cholesky calculation for a symmetric interval matrix $\mathcal{G}$ produces pivot intervals*

$$D_1, \dots, D_m$$

such that

$$\inf D_i > 0 \quad \text{for every } i.$$

Then every exact real symmetric matrix represented by $\mathcal{G}$ is positive definite.

Proof. Let G be an exact real symmetric matrix represented by $\mathcal{G}$. Apply the ordinary Cholesky recursion to G in the same fixed order.

At the first step, the exact first squared diagonal entry lies in D_1 . Since

$$\inf D_1 > 0,$$

it is positive. Its square root therefore exists and lies in $\sqrt{D_1}$. Because the lower endpoint of $\sqrt{D_1}$ is strictly positive, the divisions defining the remaining entries in the first Cholesky column are valid. Outward-rounded interval arithmetic encloses all of those exact entries.

Assume inductively that the exact Cholesky entries through column $i - 1$ lie in the intervals already constructed. Interval addition, subtraction, multiplication, squaring, square root, and division preserve inclusion whenever the denominator interval excludes zero. It follows that the exact i th squared diagonal entry lies in D_i . Since

$$\inf D_i > 0,$$

that entry is positive, its square root lies in $\sqrt{D_i}$, and $\sqrt{D_i}$ is bounded away from zero. The exact off-diagonal entries in column i therefore lie in the intervals produced by the stated recursion.

The induction continues through all columns. Every exact Cholesky diagonal entry of G is positive. Hence G admits a Cholesky factorization with positive diagonal and is positive definite. $\square$

The six certificates in Appendix B have strictly positive lower endpoints at all four Cholesky steps. By Lemma 10, the exact canonical-height Gram matrix G_T is positive definite for each of the six specialized curves. Therefore

$$n^\top G_T n > 0 \quad \text{for every } 0 \neq n \in \mathbb{Z}^4,$$

which proves independence of the four specialized points.

C.5 Logical role of the computation

The certified computation is used only at the specialization

$$t = \frac{12}{5}.$$

The generic section identities are exact identities in $\mathbb{Q}(t)$ and do not depend on numerical height calculations.

Suppose, for one of the six selected families, that the four generic sections satisfied a nontrivial relation

$$n_1 Q_1 + n_2 Q_2 + n_3 Q_3 + n_4 Q_4 = O, \quad (n_1, n_2, n_3, n_4) \neq (0, 0, 0, 0).$$

All four sections and the curve have good specialization at $t = 12/5$, so the same relation would hold among the four exact rational points listed in Appendix A. Appendix B proves that those four points are independent. Therefore the assumed nontrivial relation cannot exist, and the four generic sections are independent.

The computation therefore proves the independence required for the generic rank lower bound. It does not compute the exact generic rank, and it does not give a rank upper bound for any of the six families or for the remaining fourteen triple classes.

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